Does the environment cap the low-acceleration boost? A reading-chain construction, the Gaia DR3 wide-binary control sample, SPARC rotation curves in their environmental field, and a prepared Gaia DR4 procedure

Ugo Lanciano
Independent researcher, Locri, Italy · ORCID 0009-0008-7618-9635

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Abstract

The connection between the two sectors. The dynamics of the framework (MM 2.70.2) gives linearised Einstein gravity in the weak field by construction, static strong fields and Friedmann's equations for the homogeneous model; it does not contain a_0. The low-acceleration sector uses the interpolating function of the radial acceleration relation of disk galaxies (McGaugh, Lelli and Schombert 2016), $g = g_N \cdot \nu(y)$ with $y = g_N / a_0$ and $\nu(y) = 1/(1 - e^{-\sqrt{y}})$, where g_N is the Newtonian acceleration of the baryons; the framework's choice is $a_0 = cH_0/6$ ($H_0 = 67.4$ km s $^{-1}$ Mpc $^{-1}$), and it reads ν as a sum of reading steps, $\nu = \Sigma_{n\geq 0} e^{-n\sqrt{y}}$. The two sectors are connected by a rule motivated by a premise of the author («the galaxy it sits in affects the observation GCD, which in turn determines its own field»; GCD translates the Italian MCD, which in the framework's vocabulary is a unit of reading, not the arithmetic operation) and completed by stated hypotheses: a system's effective boost g/g_N cannot exceed the boost of the context it sits in, $\min(\nu(y), \nu(e))$, with e the context's Newtonian field in units of a_0. The translations were developed with Claude, an AI assistant; the final construction of the minimum was proposed by the second AI that checked every step. The protocols and programs covered by the advance registrations were registered with their fingerprints before the computations they govern, and …

Does the boost of gravity at small accelerations have a cap set by the environment?

Student version of MM record 2.80 (DOI 10.5281/zenodo.24011120), written with an AI assistant from the archived text; the archived text remains on Zenodo.

The archived text is in English.

Ugo Lanciano · Independent researcher, Locri, Italy · ORCID 0009-0008-7618-9635 · November 2026.

How to read this text. First we explain the words you need. Then comes a summary, *In short*. Sections 1 to 6 go into the details. At the end you will find how the work was done, a glossary and the references. Numbers like 2.56 or 2.70 refer to other works by the author, archived on Zenodo. This work continues *The dynamics of MM*, that is, MM 2.70 (DOI 10.5281/zenodo.23730218). Its latest version is 2.70.2 (DOI 10.5281/zenodo.23991161).

Before we start: the words you need

Acceleration. Acceleration tells how much the velocity changes every second. The velocity changes even when only its direction changes. A star that goes round the centre of a galaxy changes direction all the time. For this reason it has an acceleration towards the centre. It is gravity that gives it this acceleration.

Newton's gravity and Einstein's. Newton gave the law of gravity that we usually use: the bigger and closer a mass is, the more it pulls. Einstein made it more precise with general relativity. When gravity is weak, Einstein's theory can be simplified. This simplified form is called *linearised*. In practice it gives almost the same results as Newton.

The problem of galaxies. In disk galaxies the stars and gas far from the centre go round too fast. With only the matter we see, Newton's gravity is not enough to keep them in orbit. The matter we see, stars and gas, is called *baryonic matter*: it is made of atoms. The answers are of two kinds:

- there is an invisible matter, *dark matter*;
- or, when the acceleration is very small, gravity is stronger than Newton's.

MOND and the scale a_0. MOND is the second answer. It is an idea of the physicist Milgrom. There is a special, very small acceleration, called a_0 (read *a-zero*). Well above a_0, Newton holds. Below a_0 gravity is stronger. QUMOND is a way of writing MOND with precise equations. It allows the computations to be done even in complicated cases.

The scale a_0 in MM. In MM the scale is $a_0 = cH_0/6$. Here c is the speed of light. H_0 is the *Hubble constant*: it tells how fast the universe is expanding today. With $H_0 = 67.4$ km s^{-1} Mpc^{-1} one gets $a_0 = 1.0914 \times 10^{-10}$ m s^{-2}. The Mpc, *megaparsec*, is a distance of more than three million light years. The factor 10^{-10} means one tenth of a billionth. So a_0 is a very small acceleration. Every second, the velocity grows by about one tenth of a billionth of a metre per second.

The boost ν. In this text g_N is the acceleration that the baryonic matter alone would give, with Newton's law. g is the observed, or predicted, acceleration. The *boost* is the ratio between the two: ν = g/g_N. The letter ν is read *nu*. The boost tells how many times the real gravity exceeds Newton's. If ν = 1, Newton is enough.

The galaxy rule. In disk galaxies an observed law holds: the *radial acceleration relation*, or RAR. It was described by McGaugh, Lelli and Schombert in 2016. It is written with a number $y = g_N/a_0$. The number y tells how strong the field is compared with a_0. The rule is $g = g_N \cdot \nu(y)$, with $\nu(y) = 1/(1 - e^{(-\sqrt{y})})$.

- √y is the square root of y.
- In this formula e is a special number of mathematics, a little less than three. The term $e^{(-\sqrt{y})}$ becomes very small when y is large.
- When y is large, that is, the acceleration is well above a_0, ν(y) is almost 1: Newton holds.
- When y is small, ν(y) is large: gravity is boosted.

The environmental field. Every system sits inside a larger system. A pair of stars sits inside the Milky Way, our galaxy. A small galaxy can sit near a large galaxy. The larger system pulls the whole small system with its gravitational field. We call it the *environmental field*, or *external field*. In the rest of the text the letter e, on its own, stands for this field, measured in units of a_0. Be careful not to confuse it with the special number e of the formulas.

Wide binaries and Gaia. A *binary* is a pair of stars that go round each other. In *wide binaries* the two stars are very far from each other: hundreds or thousands of times the distance between the Earth and the Sun. This distance is called the *astronomical unit*, au. So far apart, the two stars pull on each other with a very small acceleration. Gaia is a European satellite. It measures with great precision the position and the motion of a great many stars. Its data come out in stages: DR3 is the third release (*Data Release 3*), DR4 will be the fourth.

Rotation curves and SPARC. The *rotation curve* of a galaxy tells how fast stars and gas go round at each distance from the centre. SPARC is a catalogue of 175 disk galaxies with measured rotation curves (Lelli, McGaugh and Schombert 2016). *Dwarf* galaxies are small galaxies.

Errors, σ and thresholds. Every measurement has an error. It is written with the sign ±. For example +2.7 % ± 1.3 % means: about +2.7 %, with an uncertainty of about 1.3 %. The letter σ (read *sigma*) stands for the *standard deviation*, the typical measure of the error. A difference of 2.1σ is 2.1 times its typical error. The larger the number, the less likely it is that the difference comes from chance alone. A *threshold* is a value fixed beforehand: above the threshold the result counts, below it does not. A *95 % interval* is a range of values built in a precise way: if the measurement were repeated many times, the range would contain the true value in 95 % of cases.

Median and weighted mean. The *median* is the value in the middle: half of the cases are below, half above. In the *weighted mean* some cases count more than others, for example those measured better.

Logarithms and dex. The base-ten *logarithm* of a number tells how many times you must multiply ten by itself to get it. The *dex* is its unit: one dex means a factor of ten. A few hundredths of a dex mean differences of a few per cent.

Resampling. To know how uncertain a result is, you can redo the computation many times. Each time you draw at random a new sample from the same data, putting back what you have drawn. How much the result changes from one computation to the next gives its error. This is *resampling*, also called *bootstrap*.

Monte Carlo simulation. A *Monte Carlo simulation* is a computation done on a computer with random numbers, like throwing a die a great many times. The name comes from the casino of Monte Carlo.

The words of the MM framework. Mathematical Materiality, MM for short, is the author's framework of ideas.

- A *premise* is a starting sentence of the author: a fixed point, like a postulate.
- The word *reading* has two uses. The first: a reading is an interpretation, written so that it can be put to the test. For example, T_max and Newton's law are two readings of the wide binaries. The second, in MM's vocabulary: a reading is also a unit that is counted. In this sense the boost is a sum of "reading steps", it is represented by a chain of readings, and the observation counts the common readings.
- In school mathematics, the *MCD* is the greatest common divisor. In MM's vocabulary it stands instead for a unit of reading.
- A *prescription* is a precise rule for doing the computations: it says which formula to use.
- *Registered* means written down and fixed with its date before doing the computations. In this way it cannot be changed afterwards.
- An *inconclusive* test is a test that, with today's data, cannot tell which reading is right.

In short

Two parts of MM. The dynamics of MM (MM 2.70-2.70.2) describes how bodies move. It gives three things:

- linearised Einstein gravity in the weak field, by construction;
- static strong fields, like those of neutron stars;
- Friedmann's equations for the universe taken as a whole.

But the dynamics does not contain the scale a_0. Small accelerations are a separate sector. There MM uses the galaxy rule, with $a_0 = cH_0/6$. In the MM framework the boost ν is a sum of "reading steps".

The rule that connects the two parts. It is called T_max and dates from October 2026. It says that the boost of a system cannot exceed the boost of the environment it sits in. As a formula: ν_eff = min(ν(y), ν(e)). Here "min" means: the smaller of the two. The environment acts as a *cap*.

Watch the direction. The boost is large when the field is weak. So an environment with a strong field gives a low cap, and an environment with a weak field gives a high cap.

The rule is motivated by a premise of the author. It is completed by stated hypotheses.

What this work adds, in three points.

1. **Why exactly the smaller of the two.** The work represents the boost as the mean length of a chain of readings. The system and the environment each have their own chain. If at each step the two chains use the same random draw, the mean number of common readings is exactly the minimum. This follows from stated hypotheses, not from the premise alone (section 2).

1. **A check on the Gaia DR3 wide binaries.** In a test already published in MM 2.70 there was an excess in the control separation range, where it was not expected. Now one sees that the excess is larger for pairs very close on the sky. A cause due to the instrument is plausible, but it has not been identified (section 3).

1. **Galaxies in the field of their environment.** The registered test of the cap cannot be done: in the SPARC galaxies almost no point lies in the zone where the cap acts. A replacement test does not decide between T_max and a QUMOND-type prescription (section 4).

What is not claimed. No new adjustable constant enters the rule T_max. The tests carried out do not exclude T_max, within their limits. All the limits are in section 6.

1. The two sectors and the cap rule

The dynamics of MM

The dynamics of Mathematical Materiality is in the records MM 2.70-2.70.2, that is, in 2.70 and its later versions. It gives three things.

- Linearised Einstein gravity in the weak field. This holds *by construction*: the dynamics was built on purpose to give it.
- *Static* strong fields, that is, fields that do not change in time. One example is neutron stars, very small and very dense stars.
- Friedmann's equations for the *homogeneous* model. They are the equations of the expansion of the universe, when it is imagined the same at every point.

The dynamics, however, does not contain the acceleration scale a_0.

The low-acceleration sector

For small accelerations MM uses the rule of disk galaxies, the RAR of McGaugh, Lelli and Schombert (2016):

- g = g_N·ν(y);
- y = g_N/a_0;
- ν(y) = 1/(1 − e^(−√y)).

Here g_N is the Newtonian acceleration of the baryons, that is, the one that matter made of atoms alone would give. g is the observed one. The formula for ν is called the *interpolating function*: it passes smoothly from Newton's case to the boosted case.

The choice of MM is $a_0 = cH_0/6 = 1.0914 \times 10^{-10}$ m s^{-2}, with $H_0 = 67.4$ km s^{-1} Mpc^{-1}.

In the MM framework the boost ν is a sum of "reading steps": ν = Σ_{n≥0} e^(−n√y). The symbol Σ (read *capital sigma*) means "sum". Many terms are added, one for each n, starting from n equal to zero. The first term is 1. The others are smaller and smaller. The whole sum gives exactly the formula for ν(y) written above.

The author's premise

The rule that connects the two sectors is motivated by a premise of the author, of 28.10.2026. Here it is, in his own words:

> «la galassia in cui sta incide su MCD di osservazione che a sua volta determina il proprio campo» ("the galaxy it sits in affects the observation GCD, which in turn determines its own field")

In MM's vocabulary the MCD is the unit of reading. Put in simpler words (this is a paraphrase by the AI): the galaxy in which a system is located affects the unit with which the system is observed, and this unit sets the field of the system.

The rule T_max

The premise is completed by stated hypotheses. From them comes the rule T_max, of October 2026. It says that the effective boost of a system, ν_eff = g/g_N, cannot exceed the boost of the context it sits in:

ν_eff = min(ν(y), ν(e)).

Here y is the internal field of the system and e is the Newtonian field of the context, both in units of a_0.

Isotropy is assumed, that is, that direction does not matter. Then there are three cases.

- **y > e.** The internal field is stronger than the environmental one. The boost ν(y) is smaller than ν(e). The rule coincides with the isolated law, as if the environment were not there.
- **The Solar System.** Here the fields are very strong compared with a_0. The isolated boost is so close to 1 that the difference is negligible. The environmental cap does not act.
- **y < e.** The internal field is weaker than the environmental one. The boost stops at ν(e). This is the cap.

An example in words. Think of a very wide pair of stars, inside the Milky Way. Between the two stars the field is very weak. On its own it would give a large boost. But the pair sits in the field of the Galaxy, which is stronger. With T_max the boost of the pair does not exceed the one that the field of the Galaxy would give.

2. Why the minimum

This part of the work is called step 26c. It looks for a reason why the rule uses exactly the minimum, and not another combination of the two fields. The final construction was proposed by the verifier, a second AI assistant. It proposed it after a first version did not hold.

The boost as a chain of readings

Imagine a *chain of readings*. The chain starts with one reading. At each step it continues with probability r = e^(−√y), or it stops.

An example of our own: a loaded coin. You toss the coin. If it comes up heads, the chain goes on by one reading. If it comes up tails, the chain stops. If the coin gives heads very often, the chains are long. If it gives heads rarely, they are short.

The length of the chain changes every time, but it has a mean. This mean length is exactly the boost ν(y).

- When the field is strong, y is large and r is very small. The chain stops almost at once. The boost is close to 1, as for Newton.
- When the field is weak, y is small and r is close to 1. The chains are long. The boost is large.

The system inside its environment

The system has its chain, which continues with probability e^(−√y). The environment has its own, which continues with probability e^(−√e). The two chains start together, with one reading, and go on step by step together: they are *synchronised*. The observation of a system inside a context counts the readings *common* to the two chains.

Shared draws: the minimum comes out. Suppose that at each step there is a single new random draw, which applies to both chains. The work interprets it as "one observation". It is like throwing a single die for both. With the same die, the chain that continues more rarely always stops first, or together with the other. So the common readings are exactly those of this chain. On average they are its boost, the smaller of the two: the mean common count is exactly min(ν(y), ν(e)). Among all the chains with these probabilities, it is also the largest possible mean common count.

Independent draws: a different rule comes out. Suppose instead that each chain has its own draw, like two different dice. Now sometimes one stops first, sometimes the other. The common part continues only as long as both continue. From this comes a different rule, called TP: TP = 1/(1 − e^(−(√y+√e))).

The simulation. A Monte Carlo simulation produced 10^6 pairs of chains, that is, one million. The results agree with the two formulas, within the errors of the simulation. Note: this checks that the computer calculations are done correctly. It does not check the physical motivation.

The hypotheses

The minimum follows from stated hypotheses, not from the premise alone. There are five hypotheses:

- the boost is represented by chains of readings;
- the observation is the count of the common readings;
- the draws are shared;
- there is no dependence on orientation, that is, on direction;
- the common count multiplies g_N.

The first version, which did not hold

The first version used the arithmetic *greatest common divisor* of two counts. The greatest common divisor of two whole numbers is the largest number that divides both of them with no remainder. For example, that of twelve and eighteen is six. This version did not hold, for two reasons:

- mean counts are not necessarily whole numbers;
- if one chain is *contained* in another, it does not mean that its count *divides* the other.

This outcome hits the first construction, not the rule T_max: the rule stays, with the new construction. In MM's vocabulary, MCD remains the name of the unit of reading. But the final construction does not use the arithmetic

operation.

3. A check on wide binaries

This part is called step 26a.

The test already published

In MM 2.70 a test on the Gaia DR3 data was published. It concerns wide binaries and looks for an *anisotropy*, that is, a dependence on direction. The test looks at how the pair is oriented on the sky:

- some pairs are aligned with the direction of the Galactic Centre, projected on the sky;
- other pairs lie across it.

For each group one computes the mean velocity between the two stars of a pair, *scaled*: brought to a common measure, so that different pairs can be compared. The *contrast* compares the two groups, as a percentage.

The control separation range

The test also has a *control separation range*. These are the pairs whose two stars are separated by 450 to 1,800 au. There the tested prescriptions predict small contrasts: from about −0.6 % to +2 %. It serves to see whether the method gives strange results where none are expected. But there the measured contrast was +2.7 % ± 1.3 %.

What was found

The criterion was registered before the computation. The pairs were divided into two groups, according to how close the two stars are *on the sky*. The distance on the sky is measured as an angle, in arcseconds (″). An arcsecond is a very small angle: one sixtieth of one sixtieth of a degree.

- Pairs closer than 4″: the contrast is +6.4 % ± 2.2 %, over 5,989 pairs.
- Pairs beyond 4″: the contrast is +0.7 % ± 1.6 %, over 11,139 pairs. The 95 % interval goes from about −2.4 % to +3.8 %.

The difference between the two groups is 2.1σ. It is just above the registered threshold of 2. So, under the registered criterion, the contrast of the control separation range is larger for pairs very close on the sky.

What it means

- **It is not an independent replication.** The result is qualitatively consistent with a diagnosis of August 2026. But the data are largely the same: 98.9 % of the control pairs have their midpoint within 1″ of a pair of that sample, before its further cuts. An *independent replication* would be a measurement redone with different data.
- **The cause.** An *instrumental* cause, that is, one due to the instrument or to the way of measuring, is plausible. It has not been identified.
- **Future data.** In the DR3 data, the key sample of the test prepared for the DR4 data contains no pairs below 4″. These counts and these diagnoses must be repeated on the DR4 data.
- **The fake directions.** In place of the direction of the Galactic Centre, 1,000 random reference directions were chosen. The analysis finds no dispersion among the directions larger than a *permutation null*. The permutation null is obtained by shuffling the data at random, so that any real effect disappears: it shows how much chance can give. This check, however, does not verify the overall uncertainty of the test.

4. Galaxies in the field of their environment

This part is called step 26b.

The data

The environmental Newtonian fields come from Chae and coauthors (2021), Table 3. There are 109 SPARC galaxies. The selected sample has 94 galaxies and 1,551 points of the rotation curves.

The registered test is not valid

The registered test of the cap is not valid, because usable points are missing. The cap acts where y < e, that is, where the internal field is weaker than the environmental one. Outer points of the galaxies were needed in this zone. The sample offers no useful ones: the only two points with y < e lie at the centre of two dwarf galaxies.

For each galaxy, the smallest value of y/e among the measured points was looked at. Across the galaxies, the median of this value is 9.7. In words: usually, even at the weakest point, the internal field is about ten times the environmental one. If the lowest tenth and the highest tenth of the galaxies are left out (the 10-90 % range), the values go from 2.9 to 45.

The replacement test

In place of the registered test, a replacement test was done. It was registered after the counts above and before looking at any *residual*. The residual is the difference between the measured value and the prediction.

Here is how it works.

- For each galaxy a number S is computed. S is the mean logarithmic residual from the law without environmental effect, in the zone y < 20e. From this, the same mean in the zone 20e ≤ y < 200e is subtracted. In words: the part of the galaxy closest to the environmental field is compared with a part where the environment weighs even less.
- S is computed for the data (S_obs) and for each prescription (S_pred).
- D is the mean of S_obs − S_pred over the 30 admitted galaxies. Each galaxy has the same weight. D is measured in dex. Its error comes from resampling the galaxies.

If a prescription describes the data well, D is close to zero.

The results

- **T_max.** In these 30 galaxies every point has y > e. So here T_max coincides exactly with the law without environmental effect. The result is D = −0.001 ± 0.017, very close to zero.
- **One-dimensional QUMOND.** It is a simplified version of QUMOND, with the fields all in the same direction (in the glossary it is called MQB). The result is D = +0.050 ± 0.019. It is a *tension* of 2.7 resampling errors: the result moves away from zero by 2.7 times its error. The registered threshold was 3. So the main result is inconclusive under the registered criterion.
- **Other ways of summarising.** The median and the weighted mean suggest an *intermediate suppression*: the environment suppresses the boost, but less than with one-dimensional QUMOND. Two descriptive variants, that is, computations done in a slightly different way only to describe, exceed the threshold.

The candidates

In one variant the environmental fields are raised by 1σ, that is, by an amount equal to their error. Then four gas-rich dwarf galaxies become candidates for probing the cap: D564-8, DDO154, DDO170 and NGC3741.

5. The new data from the Gaia satellite

The Gaia satellite will publish its fourth data release, DR4. It is expected on 2 December 2026. The work prepares three things.

- **An execution procedure,** prepared for the working environment that the author already uses. It runs, unchanged, the rules already published in MM 2.56, 2.60 and 2.70. It uses their program libraries and Git Bash, a program for giving commands to the computer. Some checks are done by hand.
- **A readiness check** of the registered programs, of the libraries and of the input data.
- **A descriptive addition,** registered before the data.

DR4 remeasures the same pairs of stars. For this reason it is not automatically an independent replication of DR3.

6. What is not claimed

- **No new adjustable constant** enters the rule T_max of October 2026. An *adjustable constant* is a number chosen on purpose to make the data fit. The work adds a construction, which holds if the stated hypotheses hold, and new numerical checks.

- **Gaia DR4 is one of many checks.** Its outcome bears on the readings of the wide binaries: T_max, the chain of MM 2.56, Newton. It does not bear on the framework as a whole.
- **T_max is not excluded.** The tests reported here do not exclude T_max, within their stated limits.
- **The limits of the SPARC comparison.** It does not test the cap itself. It does not choose among T_max, T_somma and T_quad (they are in the glossary). It does not test whole rotation curves.
- **What could decide.** Data near the *transition* could distinguish the prescriptions. But this needs the models and protocols described in the final notes. For the binaries, first of all, a two-body law is needed.

How the work was done

This section collects the procedure. In the archived text it corresponds to the heading and to the sections on the contents, on reproduction and on the declared exceptions.

Who did what. The framework and the premise are the author's. The work was written with an AI assistant (Claude). Every step was then checked by a second AI, Codex ("Astra"). Astra also proposed the final construction of step 26c. There was no human *referee*, that is, an expert who checks a work before publication.

The working names. In the working records this material is called "passo 26" (folder `26_strato_a0`). During its preparation the record was numbered 2.82. The name survives in some registered files, for example `genera_finto_26a.py` and the folder `05_deposito_282`.

The licences. Original text and materials: CC BY 4.0. Original code: MIT licence. The four third-party files listed in `dati/MANIFEST_SHA256.txt` are not included: `scarica_dati.py` downloads and checks them. The derived samples used by the analyses are included. Working language of the records: Italian.

What is in the package (folder `G_small_acceleration_layer`).

- `00_inventario`: what is closed, what is open, what can be done before DR4 (INVENTARIO_STRATO_A0_v2.md).
- `01_controllo_dr3`: step 26a, the DR3 control sample (NOTA_ESITO_26A_r1.md).
- `02_pronti_dr4`: the DR4 procedure and the readiness check (PROCEDURA_DR4.md, CORREZIONE_2_PREPARAZIONE_DR4.md). The log of the check is `02_pronti_dr4/PRONTI_20261106_c1.log`.
- `03_galassie_campo_esterno`: step 26b, SPARC with the environmental fields (NOTA_ESITO_26B_r1.md).
- `04_chiusura_minimo`: step 26c, why the minimum (NOTA_ESITO_26C_r2.md).
- In the main folder, the syntheses in Italian: CHIUSURA_CONCETTUALE_STRATO_A0.md and RIASSUNTO_PER_UGO.md.
- The verifier's reports (`verifica_g1`, `verifica_g2`) and the answers to each point.
- The author's premise, in `premises/`.

The registers. Each folder has its `REGISTRO.txt`. Protocols and programs were registered before the computation they govern, with their SHA-256 fingerprint and the UTC time. A SHA-256 *fingerprint* is a code computed from the contents of a file: if the file changes even slightly, the fingerprint changes. The outcomes were verified afterwards. The registered files were not changed: the corrections are separate files.

How to redo the computations.

- You need Python 3 with numpy and scipy. The author's configuration is Python 3.13, numpy 2.4.6, scipy 1.18.0.
- `python scarica_dati.py` downloads SPARC (astroweb.case.edu/SPARC), Chae et al. 2021 (arXiv e-print) and the El-Badry catalogue (VizieR TAP). It checks them with SHA-256. The option `--no-elbadry` skips the 138 MB catalogue. The option `--from-folder DIR` copies local files instead.
- `python riproduci.py` re-runs the computations of steps 26a-26c in a temporary copy. It takes a few minutes.
- With all four third-party inputs, 15 runs are made. Without the El-Badry catalogue, the rebuilding of the step-26a sample is skipped: the shipped sample is used and 14 runs are made.
- Before each block, `riproduci.py` checks the inputs. At the end it writes one of three messages:
- «COMPLETE: 15 of 15 runs passed»: full reproduction;
- «PARTIAL: n of 15 runs made, all passed»: some inputs were missing;
- «SOME RUNS FAILED OR DIFFER»: at least one run failed or differs.

The comparison criteria. They are listed run by run in the *docstring* of `riproduci.py`, the description written at the start of the program. In short:

- byte for byte: the log files and JSON files of steps 26b and 26c, the rebuilt samples and the shuffled sample (by its SHA-256);
- line by line, with line endings normalised: the printouts and the analysis logs of step 26a, without the lines containing paths and without the line of the overlap with August;
- the analysis JSON of step 26a: without the overlap field and the path, with the numbers to 10^{-9} in relative terms. Matrix products change in the last digits with the number of *threads*, that is, of the computations done in parallel;
- the count of the overlap with August is not reproduced: the analysis is re-run without the August 2026 file, which serves only for that descriptive count;
- the two truncated reference logs: compared on their 14 lines;
- not compared: the JSON files of the tests on fake data.

The acceptance test. The package was extracted from scratch, with local copies of the four third-party inputs verified with SHA-256. All 15 runs passed these criteria (`COLLAUDO_DA_ZERO.txt` of the deposit folder). The network download of `scarica_dati.py` is written, but it was not tried in that acceptance test.

What cannot be redone here. The DR4 analyses cannot be done before DR4. The readiness check refers to folders on the author's computer.

Declared exceptions and slips.

- The two reference logs of the step-26b tests on fake data were saved truncated, at 14 lines.
- In step 26c, at first the total field of the Galaxy at the Sun was used ($g_e/a_0 = 1.966$). The Newtonian one was needed instead ($e_N = 1.351$). The two are related by $e_N \cdot \nu(e_N) = 1.966$. The error was corrected with a registered correction, before any conclusion.
- The fake sample of the step-26a test had been produced by an unsaved piece of code. `genera_finto_26a.py` rebuilds it byte for byte: it is a declared reconstruction.
- Three files used by the DR4 rules of MM 2.60 (two orbit libraries and one module) had no recorded fingerprint before 6.11.2026. The fingerprints were taken that day (`02_pronti_dr4/IMPRONTE_SENZA_REGISTRO_20261106.txt`).
- In step 26b Claude had estimated in advance that, with $a_0 = 1.2 \times 10^{-10}$, D(M0) would shift by +0.01. The estimate was wrong: the measured shift is −0.002. Here M0 stands for the law without environmental effect.

Related records. MM 2.56 (DOI 10.5281/zenodo.23506401), MM 2.60 (DOI 10.5281/zenodo.23533882), MM 2.70 (DOI 10.5281/zenodo.23730218) with version 2.70.2 (DOI 10.5281/zenodo.23991161), MM 2.78 (DOI 10.5281/zenodo.23928201), overview (DOI 10.5281/zenodo.23535916).

This version. This student version was written with an AI assistant (Claude) from the archived text in English. First a full Italian translation was made. Then the translation was rewritten for a school student. The numbers were checked with a program: the translation against the English text, this version against the translation. The archived text is the authoritative one.

This English text is a translation of the Italian student version, which the author read and approved.

Glossary

The words of the framework (MM)

- **Cap.** The limit that the environment puts on the boost of a system, in the rule T_max.
- **Chain of readings.** A sequence of readings that starts with one reading and at each step continues with a certain probability. Its mean length is the boost.
- **MCD.** In mathematics, the greatest common divisor (in English, GCD). In MM's vocabulary it is a unit of reading; the final construction does not use the arithmetic operation.
- **Premise.** A starting sentence of the author, a fixed point like a postulate.
- **Reading.** Two uses. An interpretation, written so that it can be put to the test, like the readings of the wide binaries. Or, in MM's vocabulary, a unit that is counted: the reading steps and the chains of readings.
- **Reading step.** Each term of the sum that gives the boost ν.
- **Registered.** Written down and fixed with its date before the computations, so that it cannot be changed afterwards.
- **Shared draw.** A single random draw that applies to two chains together. The work interprets it as "one observation".

The words of physics and the symbols

- **a_0.** A very small acceleration: below a_0 gravity is stronger than Newton's. In MM $a_0 = cH_0/6$.
- **Acceleration.** How much the velocity changes every second.
- **Anisotropy.** Dependence on direction.
- **Arcsecond (″).** A very small angle: one sixtieth of one sixtieth of a degree.
- **au.** The astronomical unit, the distance between the Earth and the Sun.
- **Baryonic matter.** Matter made of atoms: stars and gas.
- **Boost (ν, ν_eff).** How many times gravity exceeds Newton's: ν_eff = g/g_N. ν(y) is the boost of the isolated law.
- **Dark matter.** An invisible matter that would account for the extra mass in galaxies.
- **Dex.** The unit of base-ten logarithms: one dex is a factor of ten.
- **g_N, g.** The Newtonian acceleration of the baryons; the observed, or predicted, acceleration.
- **Gaia DR3, DR4.** The third and the fourth data release of the Gaia satellite.
- **Isotropy.** The fact that direction does not matter.
- **M0.** The law without environmental effect.
- **Median.** The value in the middle: half of the cases below, half above.
- **MOND.** Milgrom's idea: below a_0 gravity is stronger than Newton's.
- **MQB.** The one-dimensional QUMOND prescription, with aligned fields and the same ν and a_0: $g/a_0 = \nu(y+e)(y+e) - \nu(e)e$. With the "simple" function it is exactly eq. 2 of Chae and coauthors 2021. It is not QUMOND in general.
- **Permutation null.** What you get by shuffling the data at random, so that any real effect disappears: it shows how much chance can give.
- **Prescription.** A precise rule for doing the computations.
- **QUMOND.** A way of writing MOND with precise equations.
- **r_M, x_p.** r_M is the MOND radius $\sqrt{(GM/a_0)}$ of a pair, with M the total mass of the two stars. It is the distance at which the relative acceleration of the two stars according to Newton, GM/r^2, is equal to a_0. x_p is the separation projected on the sky, in units of r_M.
- **RAR.** The radial acceleration relation of disk galaxies: $g = g_N \cdot \nu(y)$.
- **Resampling.** Redoing a computation many times, drawing at random from the same data, to estimate its error.
- **Rotation curve.** The speed at which stars and gas go round at each distance from the centre of a galaxy.
- **σ (sigma).** The standard deviation, the typical measure of the error.
- **SPARC.** A catalogue of 175 disk galaxies with their rotation curves.
- **T_max.** The cap rule: ν_eff = min(ν(y), ν(e)).
- **T_somma, T_quad.** Other possible rules: ν((y^p + e^p)^(1/p)) with p = 1, 2.
- **TP.** The rule that comes from independent draws: 1/(1 − e^(−(√y+√e))).
- **Wide binary.** A pair of stars bound by gravity, very far from each other.
- **y, e.** The internal Newtonian field and the environmental one, in units of a_0.

References

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- MM records:
- MM 2.56 (10.5281/zenodo.23506401);
- MM 2.60 (10.5281/zenodo.23533882);
- MM 2.70 (10.5281/zenodo.23730218), version 2.70.2 (10.5281/zenodo.23991161);
- MM 2.78 (10.5281/zenodo.23928201);
- overview (10.5281/zenodo.23535916).